

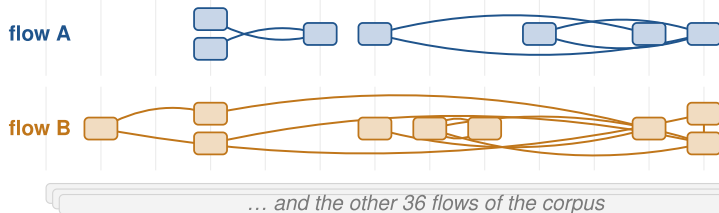
ATT&CK tactics, kill-chain order

Reconnaissance
Resource development
Initial access
Execution
Persistence
Privilege escalation
Stealth
Defense impairment
Credential access
Discovery
Lateral movement
Collection
Command and control
Exfiltration
Impact

L0 Campaign intelligence

one attack flow per incident

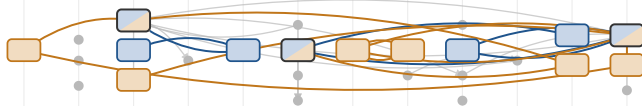
- attack flow A
- attack flow B
- in both flows
- other flows



aggregate

L1 Attack graph

every flow, one graph



condition on objective

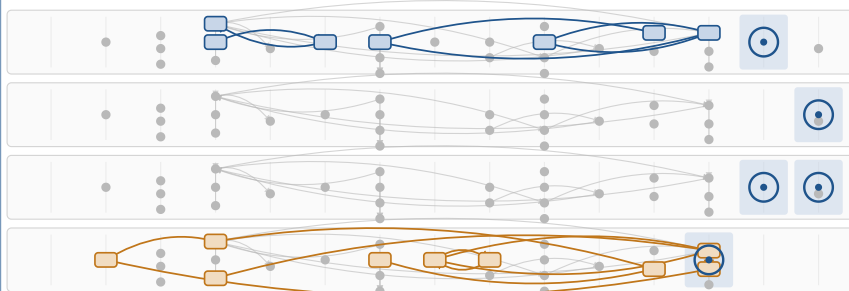
L2 Attack profiles

Exfiltration objective

Impact objective

Double extortion

No realised objective

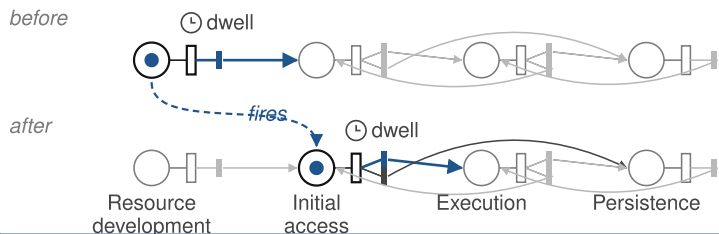


give executable semantics

L3 Petri net

one profile, one net

- tactic (place)
- timed dwell
- weighted move
- token



traverse

L4 Traversal in MTDSim

the runtime loop

